Python Programming Fundamentals

Driver and Main Function

Programming Fundamentals Team

SETU

2026-06-12

Outline

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1.1 Separating Concerns

In OOP, we separate:

- **Model classes** — `Product`, `Shop` (define data + behaviour)
- **Driver** — the file that **uses** those classes (the "main" program)

Benefits:

- Classes stay reusable across multiple programs
- Driver is easy to read — it tells the story of the program
- Easier to test each part independently

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2.1 Writing a clean entry point

```python
# driver.py
from product import Product

def main():
    apple = Product("Apple", 1.99, 101)
    bread = Product("Bread", 2.50, 102)
    milk  = Product("Milk",  1.20, 103)

    print(apple)
    print(bread)
    print(milk)

    # Update a product
    apple.set_price(2.29)
    print(f"Updated price: {apple}")
```

2.1 Writing a clean entry point

```
if __name__ == "__main__":
    main()
```

Outline

3.1 Using from ... import

```
from product import Product
```

- product is the **file name** (without .py)
- Product is the **class name** inside that file
- Both files must be in the **same folder**

File structure:

```
shop_app/
├── product.py    ← defines the Product class
└── driver.py     ← imports and uses Product
```

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4. Creating Products in main()

```python
def main():
    # Create several Product instances
    products = [
        Product("Apple",  1.99, 101),
        Product("Bread",  2.50, 102),
        Product("Milk",   1.20, 103),
        Product("Butter", 3.00, 104),
    ]

    # Print each product
    for p in products:
        print(p)

    print(f"\nTotal products: {len(products)}")
```

Output:

4. Creating Products in main()

```
Product [101]: Apple – €1.99
Product [102]: Bread – €2.50
...
```

Outline

5.1 Why do we need it?

```
if __name__ == "__main__":
    main()
```

- When Python runs a file directly, `__name__` equals `"__main__"`
- When a file is **imported** by another module, `__name__` equals the module name
- The guard ensures `main()` only runs when the file is run **directly**

Without the guard: importing `driver.py` elsewhere would run `main()` immediately — almost always unwanted.

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6. Calling Methods on Objects

```
def main():
    apple = Product("Apple", 1.99, 101)

    # Read attribute
    print(apple.get_name())     # Apple
    print(apple.get_price())    # 1.99
    print(apple.get_product_id())  # 101

    # Update attribute
    apple.set_price(2.49)
    apple.set_name("Granny Smith Apple")

    print(apple)
    # Product [101]: Granny Smith Apple – €2.49
```

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7.1 Formatted output

```
def print_product_details(product: Product):
    print("=" * 40)
    print(f"  ID    : {product.get_product_id()}")
    print(f"  Name  : {product.get_name()}")
    print(f"  Price : €{product.get_price():.2f}")
    print("=" * 40)

def main():
    apple = Product("Apple", 1.99, 101)
    print_product_details(apple)
```

Output:

```
========================================
  ID    : 101
  Name  : Apple
```

7.1 Formatted output

```
  Price : €1.99
========================================
```

Thanks for Watching - Any questions?